

Face Recognition-Based Attendance System

This presentation introduces a project that automates attendance. It uses face recognition to improve accuracy and efficiency.

Our goal is to leverage technology to create a streamlined and reliable system.

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The Problem: Inefficiencies of Manual Attendance

Time-Consuming

Manual attendance wastes class time, about 10-15 minutes per session.

Susceptible to Fraud

Proxy attendance leads to inaccurate records, with up to 20% error rates.

Resource Intensive

It requires dedicated personnel and physical resources.

Lacks Real-Time Insights

Delayed data hinders timely intervention.



Objectives: Automating Attendance for Efficiency and Accuracy

Automate

Replace manual processes with a fully automated system.

Minimize Proxy

Use face recognition to ensure accurate identification.

Real-Time Tracking

Provide real-time data for monitoring.

Contactless

Offer a hygienic attendance method.



Technology: Face Recognition – How It Works

Facial Feature Extraction

Identifies unique facial landmarks, like distance between eyes (80 nodal points).

Enrollment Process

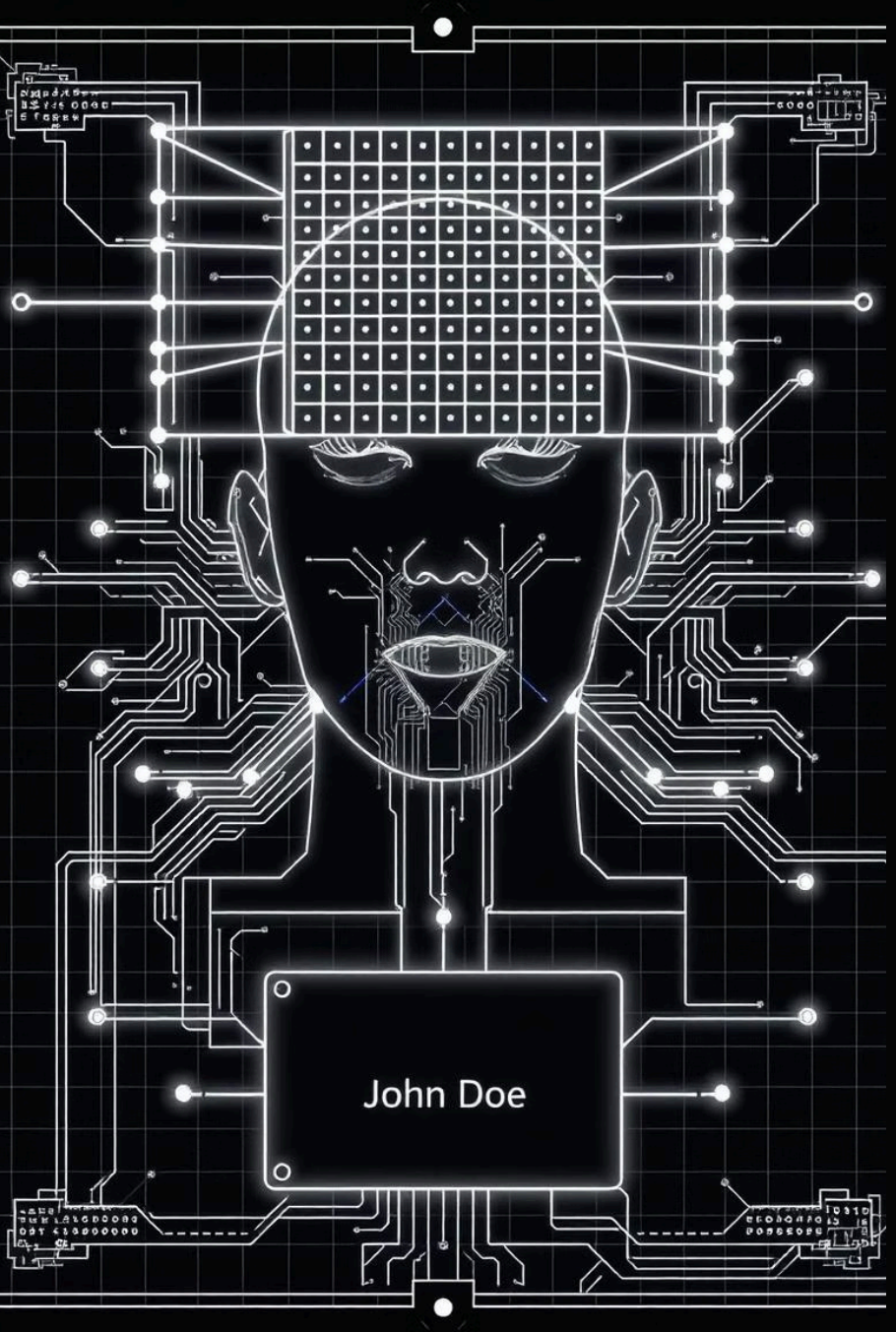
Students enroll by providing images stored as unique facial signatures.

Matching Algorithm

Compares captured data against enrolled profiles. Accuracy exceeds 99%.

Liveness Detection

Anti-spoofing measures prevent fraud using photos or videos.



System Architecture

1

Camera Module

Captures facial images at entry/exit points (1080p+ resolution).

2

Processing Unit

Handles real-time face detection and recognition (GPU-accelerated).

3

Database

Stores facial signatures and attendance records (encrypted storage).

4

Software Application

User-friendly interface for monitoring (web-based dashboard).

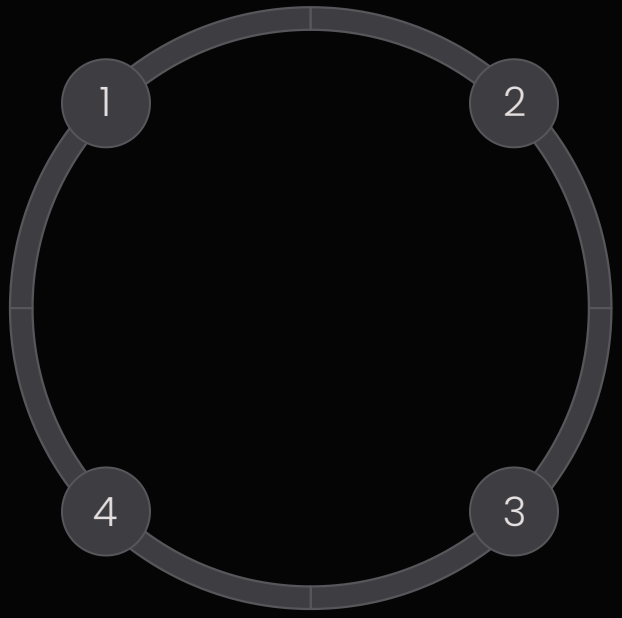
Benefits and Advantages

Increased Accuracy

Significantly reduces errors compared to manual systems (potential reduction to less than 1%).

Real-Time Monitoring

Enables immediate access to attendance data.



Improved Efficiency

Saves time and resources (estimated time savings of 5-10 minutes per session).

Contactless

Reduces disease transmission.

Implementation Plan and Future Enhancements

Pilot Program

Initial deployment in a controlled environment for testing.

Phased Rollout

Gradual expansion based on pilot results.

Integration

Connect with existing student/employee systems.

Future enhancements include access control integration and AI-powered analytics.





Conclusion: Transforming Attendance with Face Recognition

Face Recognition-Based Attendance System surpasses traditional methods.

It offers an accurate, efficient, and contactless solution.

The system improves attendance management and institutional efficiency.

Any questions?